Broadband Speckle Imaging Pipeline: Algorithm Description

Broadband_AI_solve_new.py — Mathematical and Computational Summary

Revised to match the current production source

This document describes the algorithm as implemented in `Broadband_AI_solve_new.py`. It supersedes the previous revision, which described `Broadband_AI_support.py` and no longer matches the code in several load-bearing places — the positivity map, the dispersion implementation, the matched-filter normalisation and the detection statistic have all changed, and three substantial capabilities (support-mask constraints, the MRL error metric, and companion spectral extraction) have been added. Section 0 summarises the differences; sections 1–20 are the full description; section 21 records the current switch settings and section 22 lists material from the previous revision that is now superseded or has moved elsewhere.

0. Summary of changes since the previous revision

Area	Previous revision	Current code
Frames / band	N_{frames} = 100; band quoted 400–914 nm	nim = 800; band endpoints 400 / 920 nm, realised red slice 914.3 nm (§2)
Turbulence	"von Kármán or Kolmogorov"; no zenith term	Kolmogorov only (pure $\rho^{-11/3}$, no outer scale); r_0 carries a $(\cos z)^{3/5}$ slant-path factor (§3)
Dispersion	Convolution with a delta planted at int(i_0 - ΔR); integer-quantised	Fourier shift theorem, sub-pixel exact, kernels precomputed outside the frame loop; wrap guard at dim/4 (§4)
Positivity	o = $vars^2$; "2v damping suppresses faint companions (dead-zone problem)" recorded as a known defect	φ(v) = v^2 + 2cv with c = $\sigma_{read}\sqrt{g}$ (default), or φ(v) = v. The v^2 map is **removed**: φ′(0) > 0 in both surviving modes, so the dead zone is gone by construction (§7)
Gradient chain rule	∂L/∂v = 2v · ∂L/∂o	∂L/∂v = φ′(v) · ∂L/∂o with φ′(v) = 2(v + c) (§11)
Variance floor	Hard clip of the Poisson base at zero	Smooth rectifier s(x) = ½(x + √(x² + ε²)), ε = 1 e-, so the model-based σ stays differentiable through zero (§10)
whiten = 2	IRLS described in outline	Explicit outer loop with a defined inner budget; the d(σ²)/d(model) term is deliberately absent (§12)
Support mask	not present	Optional binary FITS mask enforced through initialisation, L-BFGS-B bounds and an explicit post-solve projection; reused by every downstream diagnostic (§8)
Initialisation	flat cold start, unspecified scale	Flux-matched cold start derived from the unit-volume PSF identity, with a known-answer check on the PSF DC term (§9)
Error metric	weighted least squares only	Least squares *or* the Jefferies & Hart MRL metric, with an analytic null replacing the earlier Monte-Carlo calibration (§13)
Matched filter	Whitened per-slice residuals correlated with the frame-averaged per-slice PSF	Raw residuals in electrons correlated with the per-frame, wavelength-averaged unit-volume PSF; the variance map is computed but unused, so the filter is currently unweighted (§16)
Detection statistic	mf_coadd_hpf_wsnr = MF_{hpf} / radial noise profile σ(r)	mf_coadd_hpf_snr from a Gaussian fit to the core of the MF_{hpf} pixel histogram — a single global σ (§16)
FAP machinery	Random Field Theory (Adler) and sign-flip null described as part of the pipeline	Neither is in this file; both live in detect_sources0.py (§22)
Photometry of the companion	not present	Joint linear regression across all slices at a known position, with nuisance marginalisation and a reported covariance (§18)

Area	Previous revision	Current code
Run artefacts	FITS outputs only	Timestamped run directory with a tee'd log, PNG figures, run.json and a self-contained run.html report (§20)

1. Instrument and observation model

A ground-based telescope of diameter D observes a satellite through atmospheric turbulence and records broadband short-exposure speckle frames. Several quantities that read like free parameters are in fact derived, and are listed as such below — in particular the array size and the number of wavelength slices follow from the pupil sampling, not from a direct setting.

Quantity	Variable	Value	Notes
Telescope diameter	D	360 cm (3.6 m)	set directly
Central obstruction	obstruction_frac	0.0	enters the radiometry only
Band endpoints	lamb1, lamb2	400 nm, 920 nm	lamb2 fixes the minimum pupil radius only; the realised red slice is 914.3 nm
Pixel scale	pixscale	0.013980 arcsec/px	$1.22\lambda_1/(2D)$: Rayleigh/2, i.e. Nyquist at 400 nm
Screen diameter	D1	720 cm	2D
Array size	dim	512 × 512	derived: npts = D1/2 = 360, dim = $2^{\lfloor \log_2 \mathrm{npts} \rfloor + 1}$
Screen extent	Dscreen	1024 cm (2.84 D)	derived: dim·D1/npts
Frames	nim	800	was 100 in the previous revision
Wavelength slices	nslice	10	derived from the pupil-radius grid, §2
Slice spacing	incr	8 px of pupil radius	sets nslice
Fried parameter	r0_500	17 cm at 500 nm, zenith	13.01 cm at 400 nm, so $D/r_0 = 27.7$
Zenith angle	zen	0°	drives dispersion, airmass and the r_0 slant-path factor
Atmosphere	P, T, fwv	600 mmHg, 7 °C, 8 mmHg	refractivity inputs
Exposure time	t_exp_s	5 ms	
Read noise	read_noise	2 e- rms/px/frame	
Gain	gain	1 e-/count	data are held in electrons throughout
Target magnitude	target_mag	6.0, Johnson V (Vega)	
Random seed	random_seed	41	seeds both the screens and the noise stream

2. Wavelength slice grid

The bandpass is divided into channels on a grid of *pupil radius*, not of wavelength, so that each slice is sampled at the same number of pixels per diffraction width:

$$r_k = r_{\min} + k\,\Delta r, \quad \lambda_k = \lambda_1 \frac{\mathrm{dim}/4}{r_k}$$

with r_{max} = dim/4 = 128 px corresponding to λ_1 = 400 nm and Δr = incr = 8 px. The role of lamb2 is worth stating precisely, because it is easy to misread as the red edge of the band: it enters only through $r_{min} = \mathrm{int}(r_{max}\lambda_1/\lambda_2) + 1$, after which r_{min} is *re-derived* as r_{max} - nslice·Δr + Δr so the grid lands exactly on r_{max}. With lamb2 = 920 nm this gives r_{min} = 56 px, nslice = 10, and a reddest slice at 914.3 nm. Slice 0 is the reddest and slice 9 the bluest; the internal wavelength array is therefore *descending*, which matters wherever an ascending grid is required (the radiometry call sorts and un-sorts around this).

Slice	0	1	2	3	4	5	6	7	8	9
r_k (px)	56	64	72	80	88	96	104	112	120	128
λ_k (nm)	914.3	800.0	711.1	640.0	581.8	533.3	492.3	457.1	426.7	400.0

3. Atmospheric phase screen

One turbulent screen is generated per frame by filtering white complex Gaussian noise with the Kolmogorov power spectrum and transforming to the pupil plane. The per-mode variance is

$$\mathrm{Var}(\rho) = 0.023\left(\frac{2D_{\mathrm{scr}}}{r_0}\right)^{5/3}\rho^{-11/3}$$

where ρ is the radial distance in pixels from the array centre and the DC mode is zeroed. The real part of the transformed field is used as the phase. Note that this is a **pure Kolmogorov** spectrum with no outer scale: the previous revision's “von Kármán or Kolmogorov” no longer describes the code, which has no von Kármán branch. The variance map is precomputed once and JIT-compiled, since it is frame-independent.

Fried parameter and the zenith term

The Fried parameter is evaluated at the shortest wavelength along the line of sight:

$$r_0 = r_0(500)\left(\frac{\lambda_1}{500}\right)^{6/5}(\cos z)^{3/5}$$

> The $(\cos z)^{3/5}$ factor is new. The turbulence integral along a slant path is sec(z) times the vertical one and $r_0 \propto (\int C_n^2\ \mathrm{dh})^{-3/5}$, so omitting it would let an off-zenith run enjoy the benefit of atmospheric dispersion — wavelength-dependent image displacement, which is real diversity — with none of its cost in degraded seeing, making any off-zenith study optimistic. It is a no-op at the configured z = 0.

Per-slice phase and PSF

The screen is generated over the *bluest* pupil (radius $r_{\max}$). Slice k has a smaller pupil, so the screen is both spatially resampled and rescaled in amplitude by the same factor:

$$\phi_k(x) = s_k\,\phi_1(x/s_k), \quad s_k = \lambda_1/\lambda_k = r_k/r_{\max}$$

implemented as a bicubic `skimage.rescale` of the masked screen followed by multiplication by s_k and centred embedding in the dim × dim array. The amplitude factor reproduces the usual $\sigma_\varphi^2 \propto \lambda^{-2}$ scaling (equivalently $r_0 \propto \lambda^{6/5}$); the spatial resampling is what keeps the pupil-plane sampling consistent as the pupil shrinks. The PSF follows:

$$h_{k,l}(x) = \left|\mathcal{F}^{-1}\{A_k e^{i\phi_{k,l}}\}\right|^2$$

A_k is the unaberrated pupil amplitude normalised to unit L2 norm, so by Parseval $\sum_x h = 1$: every PSF has unit volume. That identity is not cosmetic — the cold-start flux scaling of §9 and the matched-filter normalisation of §16 both depend on it, and §9 checks it explicitly at runtime. PSFs are held in the corner-centred FFT convention. A per-frame D/r_0 diagnostic is printed from the variance of the realised screen.

4. Atmospheric dispersion

With disperse = 1 each slice is displaced vertically by the differential refraction relative to a 500 nm reference:

$$\Delta R_k = 206265[n(\lambda_k) - n(\lambda_{\mathrm{ref}})]\tan z \ \ [\mathrm{arcsec}], \quad \delta_k = \Delta R_k/\theta_{\mathrm{pix}}$$

with n from an Edlén-type refractivity in terms of P, T and water-vapour pressure. The **implementation has changed**. Previously the shift was applied by convolving with a delta function planted at $\mathrm{int}(i_0 - \delta_k)$, which quantised a general real displacement to the integer grid — up to half a pixel of error, which at z =

15° is roughly 17% of the spacing between adjacent slices. Since the *relative* slice offsets are precisely the wavelength diversity the reconstruction exploits, they must be sub-pixel exact. The shift theorem is now used instead, with a per-slice kernel

$$K_k(f) = \exp\left[-i2\pi\left((i_0 - \delta_k)f_y + i_0 f_x\right)\right]$$

applied to the PSF spectrum and followed by an fftshift. Because $2i_0$ = dim ≡ 0 (mod dim), the two half-array displacements cancel and the net effect is a pure shift of $-\delta_k$ pixels in y, leaving the PSF corner-centred as before. The kernels are frame-independent, so hoisting them out of the frame loop also removes nslice × nimages redundant forward FFTs. This matches the convention already used for the companion templates in §18.

Wrap guard. A Fourier shift is periodic: an over-large displacement wraps to the opposite edge rather than clipping, silently corrupting the PSF. The code prints the maximum and the across-band span of the shift and raises a ValueError if max|δ| exceeds dim/4, rather than returning a plausible-looking wrapped result.

5. Forward model and noise

Each broadband frame is the sum over slices of object convolved with PSF, computed by FFT:

$$m_l = \sum_k \mathcal{F}^{-1}\{\mathcal{F}\{o_k\} \cdot \mathcal{F}\{h_{k,l}\}\}$$

Noise is applied afterwards, from an independent but reproducible generator seeded from random_seed:

$$d_l \sim \text{Poisson}(\max(m_l, 0)) + \mathcal{N}(0, \sigma_{\text{read}})$$

Data are in electrons throughout (gain = 1). The noise-free cube is retained so the realised mean counts per frame can be checked against the radiometric target. Setting add_noise = 0 reproduces the earlier noise-free behaviour exactly, and in that case the code warns if a whitened metric is nevertheless selected, since the σ law assumes Poisson plus read noise.

6. Radiometry

The object cube is loaded from multi_wavelength_object.fits and scaled so that one frame detects the flux implied by the target magnitude:

$$F_{\text{total}} = F_0\, 10^{-0.4 m_V}\, A_{\text{tel}}\, t_{\text{exp}}\, \eta$$

with η the end-to-end throughput (atmosphere × optics × QE). The per-slice split comes from radiometry.calibrate, which is passed the object's own disk-integrated spectrum as the reflectance term, divided by a blackbody at T1. Spatial structure within each slice is untouched; only the slice normalisations change.

T1 is not a solar temperature. It is the reference temperature of a reflectance deconvolution: the object spectrum is divided by B_λ(T1) here and multiplied back by B_λ(T_sun) inside calibrate, and the call passes T_sun = T1 so the two cancel exactly. In this Stage-2 usage the value is therefore arbitrary — 4000, 5778, 6000 and 7500 K have been verified to give identical weights to 1 ULP — and it is kept at 6000 K only so results stay comparable with earlier runs. It has physical meaning only for Stage-1 grey objects, which this script never uses. Changing T1 *without* matching T_sun tilts the per-slice weights by up to 8.8% across 400–914 nm: a smooth slope that reads as a plausible reflectance rather than as a bug. The previous revision's description of this as “divided by a 6000 K blackbody (solar analogue)” invites exactly that mistake.

Airmass is passed explicitly as sec(z) so that it tracks the zenith angle; the default of 1.0 would otherwise model a zenith observation whatever z was set to. sec(z) is good to better than 0.3% out to z = 60°, beyond which the code warns that a Kasten-Young parameterisation would be needed.

Known issue, recorded in the source. The per-slice flux used to form the reflectance sums *all* pixels, including the companion delta function, so the magnitude calibration is slightly biased by the companion's flux. The effect is small while the contrast is small but grows as it approaches unity. The fix is to restrict the sum to support-mask pixels when a support mask is set.

7. Object parameterisation and positivity

This section replaces the previous revision's section 8 in full. The optimiser works on variables v; the object is o = φ(v). Positivity is enforced by two complementary mechanisms: the map φ itself, and an L-BFGS-B lower bound of 0 on every variable. The bound is load-bearing rather than belt-and-braces — φ is strictly increasing with φ(0) = 0, so o ≥ 0 if and only if v ≥ 0.

Mode	Map	Character
positivity = 1	φ(v) = v	Identity; positivity comes from the bounds alone. Retained as a known-good reference for validation.
positivity = 2 **(default)**	φ(v) = v² + 2cv, c = $\sigma_{read}\sqrt{g}$	Equivalently v = √(o + c²) - c: a square-root parameterisation on a pedestal c², with the pedestal subtracted so o = 0 remains attainable.

$$\phi(v) = v^2 + 2cv, \quad \phi'(v) = 2(v + c), \quad \phi^{-1}(o) = \frac{o}{\sqrt{o + c^2} + c}$$

The choice c = $\sigma_{read}\sqrt{g}$ is a convention rather than a tuned parameter: it is dimensionally correct (units of √e-), auto-scales with the detector, and requires no tuning. Its precise value is not critical — insensitivity has been measured over roughly three decades — but it must be positive and of the right order. The inverse is written as o/(√(o + c²) + c) rather than the algebraically identical √(o + c²) - c because the latter cancels catastrophically when o ≪ c²: at c = 2 and o ~ 10^{-8} e- it loses about eight digits, which is exactly the read-noise-limited regime the companions live in.

Why the v² map was removed

Any φ with φ′(0) = 0 — o = v², or any pure power o = v^p with p > 1 — makes v = 0 a stationary point for *any* gradient. Zeroed pixels can then never grow, warm starts from a sparse estimate stall outright, and the optimiser can halt at points that are not KKT points of the constrained problem. This is the "2v damping suppresses faint companions (dead zone)" caveat the previous revision recorded as a live defect of the production code. It is now fixed rather than tolerated: both surviving modes have φ′ bounded away from zero on v ≥ 0 (1 for mode 1, 2c for mode 2), including at the boundary. **Do not reintroduce a φ whose derivative vanishes at the origin**; the code raises on positivity values outside {1, 2} and on c = 0 for exactly this reason.

Choosing between the two modes

Both maps are diffeomorphic on [0, ∞), so they share the same feasible set, the same KKT points and the same unique constrained minimum. They differ only in the path taken to get there — and since runs stop on an iteration budget rather than at convergence, the path is what you actually get. Measured at 512² × 10 slices × 100 frames:

Mode	Iterations	Wall time	Final χ²	Mean relative RMSE
positivity = 2	50	1145 s	2.4141×10^7	63.0%
positivity = 1	170	3400 s	2.5127×10^7	80.4%

Mode 2 is better at every wavelength and is the production setting. Estimate FITS files are written in *object* space, so a run started under one setting can be warm-started under the other.

8. Support mask constraint

New in this revision. support_mask_file points at a 2D binary FITS image in display space, the same convention as satellite_mask.fits; it is y-flipped on load to reach the internal convention (§19) and must match the array size. When set, it is used in four distinct places, and the distinction matters because they fail differently:

- **Initialisation.** The cold-start flux is spread over the free pixels only. Applying the mask *after* building a uniform object — the earlier ordering — discarded the out-of-support flux and left the initial model too faint by a further factor $n_{support}$/dim².
- **Optimiser bounds.** Free pixels get (0, None); pixels outside the support get (0, 0). Pinning through the bounds rather than by projection inside the objective keeps the quasi-Newton history clean for the

active set.

- **Post-solve projection.** L-BFGS-B does not necessarily return bounded variables at exact machine zero, so the returned vector is explicitly zeroed outside the support before the object is formed.
- **Downstream.** The mask defines satellite versus companion regions for the contrast curve (§15), the RMSE mask, the matched-filter model mask (§16) and the exclusion region for spectral extraction (§18).

Flux inside and outside the support is printed at each stage; the outside figure should read zero, and is a cheap check that the convention has not been flipped.

9. Initialisation and restart

Because every PSF has unit volume (§3), the per-frame detected flux *is* the total object flux: $\sum_x d_l(x) = \sum_k \sum_x o_k(x)$. The cold start uses that identity directly, spreading the mean per-frame flux uniformly over the free pixels:

$$o_{\text{init}} = \frac{1}{n_{\text{free}}} \cdot \frac{1}{L} \sum_l \sum_x d_l(x), \quad v_{\text{init}} = \phi^{-1}(o_{\text{init}})$$

> The previous version set the variables, rather than the object, to the flux target, and summed the data over all frames instead of averaging. At dim = 128, nslice = 10, nimages = 100 and ~10^8 e- per frame that made the initial object about 6 × 10^6 times too bright. The scaling now rests entirely on the PSFs being unit-sum, so the code verifies it: FTpsf[0,0,k,l] is the DC term of the PSF, and a warning is issued if it departs from unity by more than 1 part in 10^6.

With restart = 1 the variables are instead warm-started from an estimate FITS file, read back through the inverse of the output transform (§19) and mapped through φ^{-1}. Because estimates are stored in object space this is independent of which positivity mode produced them. The estimate path is currently hard-coded near the top of the script and should be edited per run.

10. Error-metric whitening and the sigma law

All whitened metrics share one variance law:

$$\sigma_l^2(x) = \frac{s(\text{base}_l(x))}{g} + \sigma_{\text{read}}^2, \quad s(x) = \frac{1}{2}\left(x + \sqrt{x^2 + \varepsilon^2}\right)$$

The rectifier s is a C^∞ stand-in for clip(x, 0), differing from it by at most ε/2 — half a count at the default ε = 1 e-. It replaces a hard clip, and the reason is specific: a hard clip kinks the gradient wherever the model crosses zero, and that kink kills the L-BFGS-B line search. That was a measured failure mode in the companion pd_mrl2 code, not a hypothetical. The rectifier has one consequence worth carrying forward: it adds about 0.5 e- in the faint limit, so a correctly whitened residual has variance near 0.89, not 1, where the model is close to zero. Section 13 measures this rather than assuming it.

Mode	base	Behaviour
whiten = 0	—	Unweighted $\chi^2 = \sum(d - m)^2$. Reproduces the earliest metric exactly.
whiten = 1	data	Poisson variance estimated from the data frame. Weights are computed once and held **fixed** for the whole optimisation, so the objective is quadratic in the model.
whiten = 2 **(default)**	model	Statistically correct weights, but model-dependent — handled by the IRLS loop of §12. Intended for use once the model is already a fair representation of the data, e.g. on a restart.

11. Least-squares objective and gradient

$$\mathcal{L} = \sum_l \sum_x \frac{\left| d_l(x) - \sum_k (o_k \star h_{k,l})(x) \right|^2}{\sigma_l^2(x)}$$

The objective and gradient are split across two functions that share state. func() accumulates the model frame by frame, forms the residual, evaluates χ^2, and stores the per-pixel seed $g_l = \partial L/\partial m_l = -2r_l/\sigma_l^2$. dfunc() then cross-correlates that seed with the PSFs — the adjoint of the forward convolution — and applies the chain rule through φ:

$$\frac{\partial \mathcal{L}}{\partial v_k} = \phi'(v_k) \sum_l \mathcal{F}^{-1}\left\{ \mathcal{F}\{g_l\} \cdot \overline{\mathcal{F}\{h_{k,l}\}} \right\}$$

The chain-rule factor is φ′(v) = 2(v + c), *not* 2v. Under the removed v^2 map the factor vanished wherever the object was zero, which is what produced the dead zone of §7; with the pedestal it is bounded below by 2c, so no pixel can have an identically vanishing gradient purely because its current value is zero.

> **Implementation coupling.** dfunc() consumes the grad_seed left behind by the most recent func() call. Any code that probes the gradient must therefore re-evaluate func() at the base point before calling dfunc(), or it will correlate a stale seed. The finite-difference checker does this explicitly after each perturbed evaluation; anything new that touches the gradient must do the same.

Derivative validation

With nderiv = 1 the code runs a central-difference check on 25 randomly chosen in-support variables across six step sizes from 10^{-8} to 10^{-3}, reports mean and maximum relative error per step, re-tests at the best step with a per-index pass/fail table, and grades the outcome. Near-zero pixels are deliberately *included* rather than screened out: they carry real gradient in both positivity modes and are exactly the regime the companions occupy. Two boundary probes (variables set to 10^{-12}, and to 1000× nominal) run when the error is below 10^{-10}. The check validates whichever metric is active, least-squares or MRL. It is currently disabled (nderiv = 0).

12. IRLS outer loop (whiten = 2)

When the weights depend on the model the objective is no longer quadratic, and differentiating through the weights gives a gradient with an extra d(σ²)/d(model) term that is both awkward and destabilising. The code takes the IRLS route instead: freeze the weights, minimise the resulting quadratic, then recompute the weights from the new model and repeat.

- **Initialisation.** inv_var_model is computed from the starting variables *before* the first objective evaluation, so the reported initial χ^2 is already on the right weighting.
- **Schedule.** Each outer step gets niter_inner = max(10, niter//5) inner L-BFGS-B iterations; the outer loop repeats until the total budget niter is spent. An inner solve that converges early simply moves on with updated weights.
- **Gradient.** No d(σ²)/d(model) term appears. This is a deliberate consequence of the IRLS split, not an omission — within an inner solve the weights are constants, and that is what makes the inner problem well posed.
- **Reporting.** Iterations, function evaluations and the χ^2 reduction factor are accumulated across outer steps and reported as totals, since the per-call figures from scipy would otherwise only describe the last inner solve.

13. The MRL error metric

New in this revision. Setting error_metric = 'mrl' replaces least squares with the Jefferies & Hart MRL metric, which minimises the autocovariance of the whitened residuals rather than their sum of squares. The distinction is the point of the method: least squares drives residuals towards being *small*, MRL drives them towards being *white*, and structured residual left by an imperfect model is visible to the second criterion in a way it is not to the first. The σ law follows the whiten setting exactly as in §10. MRL can be run standalone from a flat start, or — more usefully — as a second stage warm-started from a converged least-squares solution.

Lag menu

Two families of lag are used. At τ = 0, half-plane spatial lags within a circular radius mrl_R (Δy > 0, or Δy = 0 and Δx > 0, which avoids double-counting conjugate pairs). At τ = 1…mrl_tmax, *zero-spatial-lag*

temporal lags only. The temporal family is there because it is noise-bias-free: static model error appears in every frame and so survives a temporal lag, whereas photon and read noise are independent between frames and average away. At the configured R = 25 and tmax = 10 this gives 980 spatial + 10 temporal = 990 lags. The analysis mask is the support dilated by mrl_psf_radius (50 px), which restricts the sums to signal-bearing pixels instead of diluting them with empty field.

$$C(\tau,\delta) = \frac{1}{L-\tau}\sum_{l}\sum_{x} M(x)\, r_l(x)\, r_{l+\tau}(x+\delta)$$

$$\mathcal{L}_{\mathrm{MRL}} = \frac{1}{N_{\mathrm{lag}}}\sum_{j}\left(\frac{C_j-\mu_{0,j}}{\sigma_{0,j}}\right)^2$$

The per-lag normalisation matters for interpretability: σ_0 scales as $\sqrt{N_{mask}}$, so without it the objective is not comparable between runs with different analysis masks or lag sets. Normalised, it reads as χ^2 per lag and should sit near 1 for white residuals.

Analytic null

The null is the distribution of C for white residuals. Because the menu excludes the zero lag, it can be written down in closed form:

$$E[C] = 0 \text{ exactly at every lag}, \quad \mathrm{Var}[C] = s^4\,\frac{N_{\mathrm{mask}}}{L-\tau}$$

C sums $(L - \tau)\cdot N_{mask}$ product terms, each of variance $E[r_0^2]E[r_1^2] = s^4$, and the terms are pairwise uncorrelated — the value $r(x_0)$ appears in two terms of a $\tau = 0$ sum, but their covariance is $E[r(x_0)^2]E[r(x_0+\delta)]E[r(x_0-\delta)] = 0$. Poisson's non-Gaussian higher moments do not enter, because $\mathrm{Var}[r_0 r_1] = E[r_0^2]E[r_1^2]$ for independent factors whatever their skewness. s^2 is *measured* rather than assumed to be 1, for the reason given in §10.

> **Why the Monte-Carlo null was replaced.** The earlier calibration resampled about the data. Where the data is negative — roughly half the pixels in a read-noise-dominated background — clipping the Poisson mean at zero gave those pixels a deterministic positive residual mean, and the resulting null was offset by around 1.5×10^4 per lag. The error was invisible while the analysis mask covered only bright pixels and dominant once it covered the field. A closed-form null cannot acquire that class of bug, is exact where 32 MC draws gave σ_0 to only about 13%, and makes $\mu_0 = 0$ the target by definition — so any departure from whiteness registers in the objective instead of being quietly absorbed into the null.

Because σ_0 depends on s^2, and s^2 depends on the whitening, the null must be recalibrated whenever the weights change — once after inv_var_model is first set and again at every IRLS outer step — and must stay fixed across an inner minimize, or the objective would move under L-BFGS-B. The objective refuses to evaluate before the null is calibrated.

Gradient and sanity traps

The gradient chains through the ACF, the whitening, the convolution and φ. Both product terms are carried with their own mask support — the first with M, the second with the shifted mask roll(M, δ) — which is what makes the gradient exact under a non-trivial analysis mask rather than merely approximately right. Two traps guard the setup: a warning if the initial objective is well below 1 per lag (residuals cannot be whiter than white, so the null or the noise model is inconsistent), and a warning if the measured s^2 falls outside (0.5, 2).

14. Optimiser configuration

L-BFGS-B through scipy.optimize.minimize with the analytic Jacobian supplied and the bounds of §8. Options: ftol = 10^{-15}, gtol = 10^{-12}, maxfun = 5 × maxiter, maxcor = 15. The tolerances are deliberately far tighter than the run will ever reach, so runs terminate on the iteration budget rather than on a convergence test. That is worth stating explicitly because it is why the choice of φ in §7 shows up as a real difference in output quality: two maps with the same minimum still differ in where they have got to when the budget runs out.

15. Reconstruction diagnostics

- **Volumes.** Per-slice and wavelength-integrated volume of truth versus estimate, with the ratio reported as an over- or under-estimate percentage.
- **RMSE.** Computed per slice both full-frame and restricted to the support mask (falling back to the truth footprint when no mask is set), plus a relative RMSE normalised by the RMS of the truth slice. Support-masked RMSE is also expressed as a percentage of the mean truth pixel. Written to rmse_vs_wavelength.txt.
- **Strehl.** Ratio of aberrated to diffraction-limited PSF peak, with the DL peak computed once from the unaberrated pupil before the main loop so the two share a normalisation. Mean and standard deviation are accumulated over frames and reported per slice; red slices sit higher because their phase variance is smaller.

$$S_{k,l} = \frac{\max(h_{k,l})}{\max(h_k^{\mathrm{DL}})}$$

- **Contrast curve (new).** C(k) = companion flux / satellite flux per slice, taken from the truth object with the support mask separating the two regions, plus the combined per-slice detection metric Strehl × Contrast. That product, not either factor alone, is what orders the slices by detectability, and it is plotted on a twin axis against wavelength.
- **Residual cubes.** The model is rebuilt from the converged estimate and both the raw residual (d - m) and the whitened residual (d - m)/σ are written per frame to FITS, with WHITEN, RDNOISE and GAIN recorded in the header so the whitening in force can be recovered later.

Minor defect worth knowing about. In the optional per-slice display block, the residual preview indexes the residual array by slice index although that array is stored per *frame*. The plot is therefore mislabelled — it shows frame k, captioned as wavelength k. It is display-only, gated behind display = 0 by default, and does not touch the numerics.

16. Matched-filter point-source detection

The intended workflow is two runs. First restart = 0, do_matched_filter = 0, niter = N, which converges and saves estimate.fits. Then restart = 1, do_matched_filter = 1, niter = 0, which regenerates the data and PSFs from the same seed, loads the estimate, skips the optimiser entirely and runs the filter. The steps:

- **1. Mask the model.** support_mask_file is used directly as the model mask; the previous revision's interactive threshold prompt is gone. Without a support mask, all nonzero estimate pixels are used. mf_use_truth = 1 substitutes object.fits, in which case the mask zeroes the companion delta function since it lies outside the support.
- **2. Subtract.** The masked estimate is reconvolved slice by slice with each frame's PSFs and subtracted, leaving noise plus any point source: $r_l = d_l - \sum_k (\text{masked estimate})_k \star h_{k,l}$.
- **3. Mean OTF.** Per frame, the wavelength-averaged OTF $H_l = (1/\text{nslice})\sum_k F\{h_{k,l}\}$, normalised to unit volume by its DC term.
- **4. Correlate and co-add.** The companion signal builds coherently while noise averages down as 1/√L.

$$\mathrm{MF}(x) = \frac{1}{L}\sum_l \mathcal{F}^{-1}\left\{\mathcal{F}\{r_l\} \cdot \overline{H_l(\nu)}\right\}$$

Changed from the previous revision. The filter now correlates the *raw* residual in electrons with a *per-frame, wavelength-averaged* unit-volume PSF. The previous description — whitened per-slice residuals correlated with a frame-averaged per-slice PSF, summed over both k and l — no longer matches the code. A model-based variance map is computed per frame inside the loop but is not consumed by the co-add, so as it stands the matched filter is **unweighted**. That is a defensible choice in a background-limited regime and a lossy one otherwise; either way it should not be described as whitened.

- **5. High-pass.** mf_coadd_hpf = MF - G_σ ∗ MF with σ = mf_smooth_sigma = 3 px — an unsharp mask that removes the satellite halo. Both the smoothed and the high-passed maps are saved.
- **6. SNR normalisation (changed).** A Gaussian is fitted by curve_fit to the core of the mf_coadd_hpf pixel histogram: 201 bins spanning ±5σ about the median, where the initial σ is the robust 1.4826 ×

MAD estimate, so the source tail is excluded from the fit. The map is then rescaled as shown below. A MAD-based fallback is used if the fit fails, and the MAD value is printed alongside the fitted one as a consistency check.

$$\mathrm{MF}_{\mathrm{hpf}} = \mathrm{MF} - G_\sigma * \mathrm{MF}, \quad \mathrm{SNR}(x) = \frac{\mathrm{MF}_{\mathrm{hpf}}(x) - \mu}{\sigma}$$

- **7. Report.** Peak SNR with its position in both internal and FITS display coordinates, graded as significant (> 5σ), marginal (3–5σ) or not significant (< 3σ), together with the peak matched-filter value in electrons.

The normalisation is now a single global σ for the whole field, where the previous revision divided by a radial noise profile σ(r). The wsnr map is no longer produced. The global fit is more robust to small-number statistics at large radius, but it assumes the high-passed map is statistically homogeneous across the field — an assumption the radial profile was there to avoid making, and one that weakens close to a bright primary.

17. V-band contrast and magnitudes

New in this revision, and computed as part of the matched-filter block. Band weights come from radiometry.filter_response('V', λ, system='johnson'), with a Gaussian fallback centred at 540 nm with 88 nm FWHM if the import fails. Summing the truth object inside and outside the support mask, weighted by those band weights, gives the satellite and companion V-band fluxes, from which the code reports the contrast, Δm = -2.5 $\log_{10}$(contrast), and the individual V magnitudes referenced to target_mag. This makes the configured contrast directly comparable with a quoted detection limit in Δm, which previously required an off-line calculation.

18. Companion spectral extraction

New in this revision, and the largest addition. The matched filter answers whether something is there; this section answers what colour it is. Given a *known* position, the companion's flux in every wavelength slice is recovered by a single joint linear regression in the residual space. Prerequisites are enforced at runtime: restart = 1, niter = 0 and a support mask.

Setup

- **Position.** Given in DS9 display coordinates (1-based, y-flipped) as taken from the matched-filter detection, converted internally as row = (dim-1) - (y-1), col = x-1.
- **Satellite model and residual.** The converged estimate is zeroed outside the support and reconvolved per frame; the residual is $r_l = d_l - M_l$. Setting cs_use_truth = 1 substitutes the in-memory truth object, which tests the regression code in isolation from reconstruction error.
- **Templates.** $T_{k,l}(x) = h_{k,l}(x - x_0)$, built by the shift theorem with the same sign convention used to generate the data. No mirroring is involved: the matched filter here is $\sum_x d(x)h(x - x_0)$ evaluated at a single x_0, so the convolution flip never enters.

The variance must be model-based. Data-based weighting correlates the inverse-variance weights with the residual noise: when the data fluctuates low the residual is negative *and* the weight is high, so negative fluctuations are systematically up-weighted and b acquires a negative bias. Using the noise-free satellite model, $\sigma^2_l = \max(M_l, 0)/g + \sigma^2_{\mathrm{read}}$, breaks that correlation. This is the one place in the pipeline where whiten = 1 is not merely suboptimal but wrong.

Aperture and satellite exclusion

The aperture is a circle of radius companion_ap_rad about the companion, minus an exclusion region for the satellite. Which region to exclude is a real choice, controlled by companion_sat_excl_frac:

- **frac = 0** excludes the object support: where the satellite's flux is.
- **frac > 0** excludes the satellite *model footprint* — the support convolved with the frame-averaged PSF, thresholded at that fraction of its peak — i.e. where the satellite's subtraction *errors* are. The footprint is computed directly rather than approximated by a dilation radius, so the excluded region carries the true anisotropy of the PSF.

The support says where the flux is, which is the right constraint in object space. But what contaminates b is the residual left by subtracting the satellite model, and that model is the support convolved with the PSF — so

its errors are smeared over the whole PSF footprint, tens of pixels wide, not confined to the support. Excluding only the support therefore removes the region where the satellite *is* while leaving the region where its subtraction errors *are*. This shows up as a large, highly significant nuisance pedestal (+1.26 × 10^3 e-/frame at 9.2σ at m_V = 11.88) and is the leading candidate for the observed R-bar ≈ 1.3 amplitude bias. At small separations the parameter cannot be opened up freely: removing the primary's halo also removes aperture the companion templates need. Sweep it and watch R-bar, the pedestal significance and the per-slice SNRs against the printed aperture pixel count. Typical trial values are 10^{-3} to 10^{-1}.

Nuisance terms

PSFs have positive sum, so an unmodelled local pedestal p in the residual couples into b_k with weight $p\sum(T_k/\sigma^2)$ — a weight that *grows* with aperture area. Left out of the model it biases every α_k in the same direction and is invisible in the output. The fix is to append frame-independent nuisance columns to the design matrix and solve for them jointly: cs_nuisance = 'none' (physical slices only), 'const' (pedestal, recommended and the current setting) or 'linear' (pedestal plus y and x gradients). Marginalisation is then automatic, because the top-left nslice block of A^{-1} already has the nuisance directions projected out, so σ_α correctly inflates to reflect what the pedestal costs.

Normal equations and solution

A and b are accumulated streaming over frames — storing the full (nslice, nimages, n_{ap}) template cube would be roughly 2.6 GB at 400 frames and a 160 px radius:

$$A_{ab} = \sum_I \sum_{x \in \mathcal{A}} \frac{T_a(x) T_b(x)}{\sigma_I^2(x)}, \quad b_a = \sum_I \sum_{x \in \mathcal{A}} \frac{r_I(x)\, T_a(x)}{\sigma_I^2(x)}$$

$$\theta = A^{-1} b, \quad \alpha_k = \theta_k, \quad \sigma_{\alpha_k} = \sqrt{(A^{-1})_{kk}}, \quad \mathrm{SNR}_k = \alpha_k / \sigma_{\alpha_k}$$

Columns are rescaled by √diag(A) before inversion. This is a pure reparameterisation — the physical α are unchanged — but it makes the reported condition number meaningful once columns with very different units are mixed. CAUTION is printed above 10^3 (significant spectral cross-talk) and WARNING above 10^6 (severe; consider Tikhonov regularisation or fewer, coarser slices). A pinv fallback covers a failed exact solve.

Outputs and diagnostics

- Per-slice table of α_k, σ_k, SNR_k and a status flag: DETECTED at |SNR| ≥ 3, otherwise upper limit or negative.
- Fitted nuisance values with their own significance, and an explicit note when the pedestal is significant — meaning that without the term it would have leaked into every α_k.
- Truth comparison per slice with the recovered/truth ratio, when a truth object is available.
- Aperture diagnostic: the radii enclosing 90%, 99% and 99.9% of $\sum T^2$, with a note when the configured aperture is more than three times the 99% radius. Because of the inverse-variance weighting, far pixels contribute almost nothing to b — an oversized aperture is mainly slow rather than harmful, but it does add exposure to subtraction systematics.
- Files: companion_spectrum.txt, companion_spectrum_covariance.txt (the full nslice × nslice marginalised covariance, not just the diagonal) and a two-panel plot of flux with error bars against truth, plus per-slice SNR.

19. FITS coordinate conventions

Internal arrays use numpy (row, col) = (y, x) with the origin at top-left. FITS and DS9 put the origin at bottom-left with the y-axis flipped. The transform applied to every output cube is unchanged from the previous revision:

$$\mathrm{FITS} = \mathrm{flip}_y(\mathrm{swapaxes}(0,2)\,\mathrm{swapaxes}(1,2)\,[\mathrm{internal}])$$

- **Inverse** (restart, matched-filter estimate load): internal = swapaxes(1,2) swapaxes(0,2) [flip_y(FITS)].
- **2D masks** (support mask, satellite mask) are y-flipped only, with no axis swap.

- **PSF cubes** are additionally fftshifted on the *spatial* axes only. Shifting the wavelength axis would silently permute the slices.
- **DS9 to internal** (1-based input, as used for the companion position): row = (dim-1) - (ds9_y - 1), col = ds9_x - 1.

20. Run artefacts

New in this revision. Every run creates a timestamped directory YYYYMMDD_HHMMSS and writes everything into it, so runs no longer overwrite one another. Standard output is tee'd to run.log, figures are saved as numbered PNGs rather than shown interactively, and the run closes by writing a run.json metrics summary and a self-contained run.html report with the images base64-embedded alongside the full log.

File	Contents
object.fits / estimate.fits	Truth and reconstructed object cubes, (wavelength, y, x)
data.fits	Simulated frames, (frame, y, x), in electrons
psf.fits / psf_mean.fits	Per-slice PSFs from the last frame; per-frame wavelength-averaged unit-volume PSF
residuals_raw.fits / residuals_whitened.fits	Post-convergence residuals per frame, with WHITEN/RDNOISE/GAIN headers
satellite_mask.fits / masked_estimate.fits	Model mask in display space and the masked estimate used by the matched filter
mf_residual.fits	Per-frame residual after satellite subtraction
mf_coadd.fits, mf_coadd_smooth.fits, mf_coadd_hpf.fits, mf_coadd_hpf_snr.fits	Matched-filter co-add, its smoothed version, the high-passed map and the map in SNR units
companion_spectrum.txt / companion_spectrum_covariance.txt	Extracted spectrum with uncertainties, and the marginalised covariance matrix
convergence_history.txt / rmse_vs_wavelength.txt	Objective per function evaluation; RMSE full-frame, support-masked and relative
run.log / run.json / run.html	Full log, machine-readable metrics, and the HTML report

21. Current configuration

The switches as the file currently stands:

Switch	Value	Switch	Value
restart	1 (warm start from estimate FITS)	positivity	2 ($v^2 + 2cv$)
niter	0 (optimiser skipped)	error_metric	'ls'
whiten	2 (model-based, IRLS)	nderiv	0
nim	800 frames	display	0
disperse	1 (no-op at zen = 0)	fitsout	1
add_noise	1	do_matched_filter	0
support_mask_file	set	companion_spectrum	1
companion position	DS9 (270, 257)	companion_ap_rad	87 px
cs_nuisance	'const' (pedestal)	companion_sat_excl_frac	0.0

Read together, these describe a **spectroscopy run, not a reconstruction run**: the script regenerates the data and PSFs from the seed, loads an already-converged estimate, performs no optimisation at all, and goes straight to companion spectral extraction. Note also that companion_sat_excl_frac is currently 0, which is the configuration §18 identifies as the leading candidate for the amplitude bias — deliberate if the run is part of that sweep, worth revisiting otherwise.

22. Superseded and relocated material

The following appeared in the previous revision and should no longer be read as describing this file.

- **The three-phase production sequence** (Phase 1: whiten = 1, cold start, 50 iterations; Phase 2: whiten = 2 IRLS, warm start, 50 iterations; Phase 3: niter = 0, mf = 1). Still a sound way to drive the script, and §12 describes the IRLS phase accurately, but the switch set now also includes positivity, error_metric and the support mask, and there is a fourth stage — spectral extraction — that the sequence does not mention.
- **Random Field Theory (Adler formula) FAP and the sign-flip null.** Neither is implemented in this file. Both belong to detect_sources0.py and should be documented with it. The in-script detection statistic is now the Gaussian-histogram SNR of §16.
- **mf_coadd_hpf_wsnr and the radial noise profile σ(r).** Removed, and replaced by the global Gaussian-fit normalisation described in §16.
- **The vars² reparameterisation and the dead-zone caveat.** Superseded by §7. The defect the caveat described is now designed out, not managed.
- **Detection-limit figures** — 5σ $\Delta m \approx 7.2$ ($m_V \approx 13.2$) at 0.25 arcsec with 100 frames, SNR $\propto 10^{0.4\Delta m}$ with $R^2 > 0.99$, and the confirmed detection at FITS (256, 269) at 5.25σ for CSO_scale = 0.1. These were measured under the removed v² positivity map, the previous matched-filter normalisation and 100 frames. The script now defaults to 800 frames with a different φ, a different filter normalisation and a different SNR definition, so all three should be treated as historical provenance and re-measured before being quoted. The √N scaling alone would move the limit by roughly 1.1 mag at 800 frames, before any of the other changes are accounted for.
- **Validation claims.** The derivative check still passes to better than 10^{-6} relative error, but it now validates the $\varphi'(v) = 2(v + c)$ chain rule rather than 2v, so the earlier pass is not evidence for the current gradient. The null-histogram, sign-flip and RFT validations refer to code outside this file.

Validation status carried forward: the finite-difference derivative check passes to better than 10^{-6} relative error for the current chain rule; PSF unit-volume is checked at runtime; dispersion shifts are guarded against Fourier wrap; the MRL null is exact by construction rather than calibrated. Detection-limit and false-alarm figures are pending re-measurement under the current configuration.